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# DESIGN AND VALIDATION OF A RESPONSIBLE ARTIFICIAL INTELLIGENCE-BASED SYSTEM FOR THE REFERRAL OF DIABETIC RETINOPATHY PATIENTS <sup>\*</sup>

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## Abstract

Diabetic Retinopathy (DR) is a leading cause of vision loss in working-age individuals. Early detection of DR can reduce the risk of vision loss by up to 95%, but a shortage of retinologists and challenges in timely examination complicate detection. Artificial Intelligence (AI) models using retinal fundus photographs (RFPs) offer a promising solution. However, adoption in clinical settings is hindered by low-quality data and biases that may lead AI systems to learn unintended features. To address these challenges, we developed RAIS-DR, a Responsible AI System for DR screening that incorporates ethical principles across the AI lifecycle. RAIS-DR integrates efficient convolutional models for preprocessing, quality assessment, and three specialized DR classification models. We evaluated RAIS-DR against the FDA-approved EyeArt system on a local dataset of 1,046 patients, unseen by both systems. RAIS-DR demonstrated significant improvements, with F1 scores increasing by 5-12%, accuracy by 6-19%, and specificity by 10-20%. Additionally, fairness metrics such as Disparate Impact and Equal Opportunity Difference indicated equitable performance across demographic subgroups, underscoring RAIS-DR's potential to reduce healthcare disparities. These results highlight RAIS-DR as a robust and ethically aligned solution for DR screening in clinical settings. The code, weights of RAIS-DR are available at <https://gitlab.com/inteligencia-gubernamental-jalisco/jalisco-retinopathy> with RAIL.

**Keywords** Diabetic Retinopathy · Responsible AI system · Fairness

## 1 Introduction

The prevalence of Diabetes Mellitus (DM) has been steadily increasing globally, with a particularly pronounced escalation in low and middle-income countries like Mexico [1, 2]. Moreover, there is a strong correlation between the population size of DM patients and Diabetic Retinopathy (DR) patients [3] indicating a need for specialists, resources, and screening programs for DR early detection [4].

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One solution is to use AI-based systems to help grade DR specifically by classifying Retinal Fundus Photographs (RFP) for DR screening [5]. AI-based DR graduation offers several potential benefits, such as increasing efficiency by maximizing clinical coverage, scaling the screening program, and helping to provide early DR detection and treatment. However, numerous examples demonstrated that AI health systems perform exceptionally well with specific curated datasets [6, 7, 8], but their performance is often limited when deployed in clinical environments [9, 10]. The main reasons for the limited performance in real-world applications are: i) the presence of low-quality data [9], ii) the model learning something unexpected or spurious features such as the background [11], and iii) the lack of large, representative labeled datasets [7]. This work introduces a novel redesigned Responsible AI System for DR Screening (RAIS-DR), which integrates ethical principles into the classic AI lifecycle to address these challenges.

The responsible principles underpinning RAIS-DR are derived from a consensus mapping of ethical principles [12]. Transforming these principles into effective and practical actions/operations is neither trivial nor a simple task that can be automated. This strategy involved new responsible stages/actions in the typical AI lifecycle. See Table 1 in Section 3 for the link between these stages, the ethical principles, and the AI lifecycle stages.

RAIS-DR was redesigned with six efficient Convolutional Neural Networks (ConvNets), incorporating workflow steps to assess image quality in real-time and enable retaking of RFPs if necessary. Each ConvNet was trained, fine-tuned, and tested using publicly available datasets, including EyePACS [13], IDRiD [14], and MESSIDOR II [15]. As noted by [10], one of the most common pitfalls in real-world AI systems is testing only on the test split from the same dataset. To address this, the models underwent retrospective clinical validation using a separate local dataset in a pilot study involving 1,046 patients, with ground truth labels determined by consensus among three retina experts. Performance was evaluated using F1-score, sensitivity, specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and accuracy.

The experimental setup focused on detecting Referable DR (RDR) and All-Cause Reference (ACR) cases. RDR was defined as severe non-proliferative or proliferative DR according to Mexican clinical guidelines [16], while ACR included RDR cases and ungradable images. This approach ensures that RAIS-DR aligns with clinical standards and addresses real-world diagnostic challenges.

Another key contribution of this paper is the comparative evaluation of two AI screening systems, including a fairness assessment. We compared RAIS-DR with the FDA-approved EyeArt system (Eyenuk, <https://www.eyenuk.com>) on both a per-patient and per-image basis. RAIS-DR outperformed EyeArt, achieving an accuracy improvement of up to 19%. To evaluate fairness, we assessed potential biases using Disparate Impact (DI) and Equal Opportunity Difference (EOD) across factors such as sex, projection type (macula-centered vs. optic nerve-centered), laterality (left vs. right eye), and age. DI measures the ratio of positive outcomes between subgroups, with values close to 1 indicating fairness, while EOD evaluates the difference in true positive rates between subgroups, with values close to zero suggesting equitable performance. The DI values ranged from 0.984 to 1.031, indicating minimal bias and balanced treatment across subgroups. Additionally, EOD values were close to zero, further confirming RAIS-DR’s fairness across all evaluated subgroups.

GradCAM [17] was also used to analyze and visualize the inference gradients of the primary (referral) model. Finally, a git repository with cloud deployment facilitates evaluation of the RAIS-DR by other retina experts.

The remainder of the paper is organized as follows: Section 2 presents the literature review. Section 3 describes the RAIS-DR and the implementation of ethical principles across the AI lifecycle. Section 4 details the training data and the local clinical dataset used for validation. Section 5 outlines the experimental setup and evaluation metrics. Finally, the results and conclusions are presented in Section 6 and Section 7, respectively.

## 2 Literature review

Previous work [6], developed and validated a DL model on two external datasets (EyePACS and Messidor II), achieving > 90% sensitivity and specificity in distinguishing Referable-DR (Moderate DR or worse per ICO grades) from Non-Referable DR. However, Beede *et al.*[9] reported that, when deployed in Thai clinics, it reduced false negatives by 23% reduction while increasing false positives by 2%. These performance issues have been attributed primarily to low-quality images, which impact gradability, a common challenge in DR screening due to other ocular conditions such as cataracts [5, 18, 19]. To address this, quality assessment models were integrated into the AI system presented here to prompt retakes of RFPs or facilitate direct patient referrals when necessary.

In [20], code-free cloud tools were used to train a DR classification model with minimal preprocessing, resizing images to  $224 \times 224$  pixels. However, external validation showed highly variable classification accuracy for Referable DR (RDR) versus Non-Referable DR (NRDR), dropping to 10-13%. This variability could be attributed to factors such as input size, background noise in retinal fundus photographs (RFP), and geometric distortions from resizing, as identified in prior work [21]. To address these issues, the proposed approach preprocesses RFPs, increases ConvNet input size to  $512 \times 512$  pixels, and includes steps to remove background noise, spurious features such as letters, and geometric distortions, thereby enhancing model robustness [22, 23]

Scheetz et. al. [24] presented an AI-assisted DR screening with 236 participants in Australia, achieving an AUC, sensitivity, and specificity of 0.92%, 96.9%, and 87.7%, respectively. In addition, the study highlighted the most important challenges, image quality, data flow, and technology adoption by clinical users. A key distinction in the presented work is the use of a larger patient dataset (1046), combined with post-training analysis of the model’s explainability and bias measurement. This study also compares RAIS-DR system performance with an external AI screening system (Eyeart).

AI system failures in healthcare are common [10, 11], frequently stemming from issues such as insufficient external validation, unreported limitations, model bias, and lack of explainability. One approach to address these challenges is to apply a responsible AI framework. However, existing frameworks, tools, and checklists for responsible AI [25, 26] are often too general, and static, lacking adaptability to specific real-world healthcare needs[27]. These tools also offer limited feedback and fail to address challenges such as the use of non-tabular data, clinical implications, and technical constraints. An iterative process, like the AI lifecycle, enables early identification and resolution of these issues, reducing failure risks and enabling regular updates based on feedback. To ensure responsible development, the RAIS-DR system includes additional stages in the AI lifecycle; see Table 1 for details.

### 3 Responsible AI System

A key objective of responsible AI systems is to address ethical, legal, technical, and societal issues in real-world applications a priority for governments, businesses, and society. However, responsible AI is complex and interdisciplinary, so reducing it to a checklist risks neglecting ongoing monitoring and prioritizing short-term gains over long-term ethics and safety. To mitigate these issues, iterative responsible actions are implemented throughout the AI lifecycle [28].

From a large set of responsible and ethical principles [12, 28]. This work focuses on the following: professional responsibility, privacy, accountability, safety, security, transparency, explainability, human control of technology, fairness and non-discrimination, and promotion of human values and well-being. Furthermore, the responsible AI system pipeline was developed following the local clinical guidelines [16] and accounts for model limitations.

These responsible stages are outlined in Table 1, along with their corresponding implemented actions, underlying principles, and related AI lifecycle stages [28]. Note that each responsible stage may correspond to more than one responsible action. An important outcome of incorporating responsible AI practices was redesigning the system pipeline, focusing on data exploration, analysis, and standardization. Additionally, fairness metrics and explainability were implemented as well. Benchmarking and exposure to external review were crucial in addressing ethical, legal, technical, and societal considerations, ensuring a more comprehensive and accountable approach to AI-based technology development.

Figure 1 illustrates the RAIS-DR pipeline. The process begins with an RFP acquisition. The background is removed in the *Processing* stage, and the image is adjusted to the appropriate aspect ratio and size. Subsequently, the *Quality Assessment* stage determines whether the image is of sufficient quality to be graded. Based on the quality assessment results, the observed anatomy, and potential presence of other eye conditions such as cataracts, a trained professional must make an on-site medical decision (MD) regarding whether to retake the RFP. Following this, three convolutional networks are used, one for patient referral classification and two for grading Diabetic Retinopathy (DR) severity according to International Council of Ophthalmology (ICO) standards. These DR grading models are essential for adhering to clinical guidelines, providing clear results to physicians and retina specialists, and supporting prioritization in patient care.

Table 2 summarizes the RAIS-DR models, classes, ConvNet backbones, and the main actions. It is important to note that we tested with at least five different models (including visual transformers) and best trade off between performance and inference time is presented at Table2. The MP model removes the background,

Table 1: AI-Responsible stages, actions, and principles implemented in the AI lifecycle.

| AI Re-sponsible stage                        | Responsible actions                                                                                                                                              | AI life-cycle stages          | Responsible principles                                                                               |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------|
| Ethical and technical review                 | Conducted iterative reviews to ensure benefit to society, ethical principles, and technical robustness.                                                          | Plan and design.              | Professional responsibility, privacy, safety and security, promotion of human values and well-being. |
|                                              | Ensured control over data use, privacy, consent, ability to restrict processing, and the rights to rectification and erasure.                                    | Collect and process data.     |                                                                                                      |
|                                              | Addressed dispute resolution and environmental responsibility.                                                                                                   | Operate and monitor.          |                                                                                                      |
| System re-design                             | Developed custom models: created new models for image quality assessment. Adapted a custom data pipeline according to clinical guidelines and model limitations. | Build and use model. Deploy.  | Accountability, Human control of technology.                                                         |
| Exploration, analysis, standardizing data    | Avoided spurious features and applied preprocessing to maintain aspect ratio.                                                                                    | Collect and process data.     | Robustness, Professional responsibility.                                                             |
| Bias analysis, explainability                | Incorporated fairness metrics into the model evaluation.                                                                                                         | Verify and validate.          | Fairness and nondiscrimination.                                                                      |
|                                              | Computed gradient maps to highlight the learned features.                                                                                                        |                               | Explainability.                                                                                      |
| Monitoring                                   | Deploying a cloud system for feedback from retina experts.                                                                                                       | Deploy, Verify, and validate. | Robustness, human control of technology.                                                             |
| Benchmarking and exposure to external review | Methods and results to peer-review<br><br>Compared performance versus FDA-approved system. Published models and responsible AI practices.                        |                               | Robustness, and transparency.                                                                        |

while the MQ and MA are models to define quality assessment and the anatomical detection of optic nerve and macula. Model M1-Referral is the most important model for classifying if the patient is referred, and finally, M2-DR grade and M3-DR grade help us to do fine-grain DR-level classification. The entire process takes approximately two seconds to execute all models on a CPU cloud server.

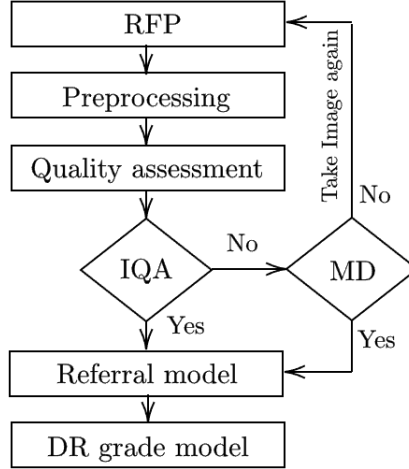


Figure 1: The RFP is the system input, which undergoes preprocessing to remove the background. Then, quality models evaluate the image in general, and an additional model detects the macula and the optic nerve. If the image is not gradable (low quality), the IQA=Image Quality Assessment and an MD=Medical Decision define if the image should be retaken.

Table 2: Summary of RAIS-DR Models, backbones, and main actions.

| Model       | Classes                                         | Backbone               | Action                                                              |
|-------------|-------------------------------------------------|------------------------|---------------------------------------------------------------------|
| MP          | 0: Background<br>1: Eye                         | Unet [29]              | Remove Background                                                   |
| MQ          | 0: Bad quality<br>1: God quality                | MobileNet [30]         | Evaluate Quality                                                    |
| MA          | 0: Macula detection<br>1: Optic nerve detection | Faster R-CNN [31]      | Anatomical detection<br>Repeat or M1                                |
| M1-Referral | 0: {R0,R1,R2}<br>1: {R3,R4}                     | EfficientNet V2-s [32] | Evaluate M2<br>Evaluate M3                                          |
| M2-DR grade | 0: {R0,R1}<br>1: {R2}                           | EfficientNet V2-s [32] | Primary care review in 12 months<br>Primary care review in 6 months |
| M3-DR grade | 0: {R3}<br>1: {R4}                              | EfficientNet V2-s [32] | Referral to retina specialist.                                      |

## 4 Data and labels

All the RFP were labeled according to the ICO severity scale: no apparent DR (R0), mild non-proliferative DR (R1), moderate non-proliferative DR (R2), severe non-proliferative DR (R3), proliferative DR (R4). In addition, for the local data, (R5) is defined as eye enucleation, and (R6) is assigned to ungradable due to inadequate image quality to confirm the grade of DR.

### 4.1 Public data: for training

Three public datasets were used to train our models: EyePACS [13], IDRiD [14], and MESSIDOR II [15]. Most of the RFP in these datasets are 45-degree field, macula centered, with different spatial resolutions, aspect ratios, cameras, and image quality. No additional demographic information is available for these public datasets.

#### 4.2 Local data: only for external clinical validation

A local dataset was gathered from three primary care facilities in Jalisco, **La Aurora y la Esperanza**, **Yugoslavia**, and **Paraísos del Colli** with the support from the IDB Fair-LAC program between August 2021 and March 2022. These clinics use three Topcon non-mydiatric cameras TRC-NW400.

The inclusion criteria for sample selection included patients with DM type II, both male, and female aged eighteen years and above, from the Cardio-metabólicas program. Two RFP one macula-centered and one optic nerve-centered were captured per eye. Mydriasis was used during this RFP acquisition for at least 498 patients. For more information about the data collected see [33]. The ethical protocol was registered with the ID number 9/E-JAL/2021 by the Secretaria de Salud de Jalisco. Patients choosing to participate were required to sign an informed consent. This study relied exclusively on anonymous data. The RFP was labeled by three selected Colegio de Oftalmólogos de Jalisco members. Table 3 presents the main local data characteristics, demographics, image quality distribution, and disease severity distribution according to ICO.

Table 3: Summary of data characteristics, demographic information, image quality distribution, and DR grade distribution of the local data for external validation.

| Characteristics                                                                                        | Local Data Set  |
|--------------------------------------------------------------------------------------------------------|-----------------|
| No. of images                                                                                          | 4296            |
| No. of optic nerve-centered images                                                                     | 2148            |
| No. of macula-centered images                                                                          | 2148            |
| No. of ophthalmologists                                                                                | 3               |
| No. of grades per image                                                                                | 3               |
| Patient demographics                                                                                   |                 |
| No. of unique individuals                                                                              | 1074            |
| Age, mean (SD)                                                                                         | 60.4 (12.1)     |
| Female, No./total (%) among patients for which sex was know                                            | 688/1046 (65.7) |
| Image quality distribution                                                                             |                 |
| Fully gradable, No./total (%) among patients for which image quality was assessed                      | 797/1046 (76.2) |
| No image (patients)                                                                                    | 28              |
| Disease severity distribution classified by majority decision of ophthalmologists (reference standard) |                 |
| Total patients for which both diabetic retinopathy and diabetic macular edema were assessed, No. (%)   | 797 (100)       |
| R0. No diabetic retinopathy                                                                            | 679             |
| R1. Mild diabetic retinopathy                                                                          | 64              |
| R2. Moderate diabetic retinopathy                                                                      | 28              |
| R3. Severe diabetic retinopathy                                                                        | 26              |
| Referable diabetic macular edema                                                                       | 0               |
| (R3+R4) Referable diabetic retinopathy                                                                 | 54              |

## 5 Experimental setup

All the models were trained with data augmentation (horizontal mirroring, rotation, blur, class balancing, and transfer learning (ImageNet pre-trained model)). The most relevant modification in the model architecture is the input shape (512x512x3) and the last layer to make a binary model. We use an IBM AC922 and an Nvidia DGX A-100 640 server to train and fine-tune the models and the PyTorch framework to implement the models.

We evaluated different clinical scenarios. First, we assessed the performance of Referable Diabetic Retinopathy (RDR) detection per patient, excluding images that were not fully gradable. According to the Mexican DR guidelines, an RDR case is defined as either severe non-proliferative DR (R3) or proliferative DR (R4) [16]. Next, we presented the All-Cause Referable Cases (ACR) classification results per patient (including images that were not fully gradable). ACR is defined as severe non-proliferative DR (R3), proliferative DR (R4), or ungradable image (R6). Performance per image was evaluated for both RDR and ACR classifications, comparing it with the FDA-approved model (Eyear) in each case.

System's performance was further examined for bias across different subgroups, including sex, eye projection, age, and laterality. The Disparate Impact  $DI$ , was used to compare the favorable outcomes for a monitored group against those for a reference group. In  $DI$  interpretation, usually when  $1 < DI < 0.8$  (known as the *four-fifths rule*), the model does not have a bias for a group. Equal Opportunity Difference ( $EOD$ ) quantifies the deviation from equality of opportunity, ensuring that the same proportion of favorable results is reached across groups. Ideally, this metric must be equal zero.

Table 4 the experimental setup cases. The first two experiments simulate a clinical environment, focusing on patient referral classification. Experiments 3 and 4 assess the model's referral performance on individual images. Finally, experiments 5 through 9 are designed to evaluate potential biases in the model. The abbreviations used in Table 4 are defined as following: Referable Diabetic Retinopathy (RDR) Class 0 as "Not referral" (R0,R1,R2) and class 1 as DR referral (R3,R4). All-cause referable (ACR) defines Class 0 as "Not referral" (R0,R1,R2) and Class 1 as "referral" (R3,R4,R6). Moreover, the projection centered on the macula is defined in the table as Proj A. The projection centered on the optic nerve is defined as B. "Both" refers to both sex: male (M) and Female (F) patients. "L" is Left , and "R" is Right eye.

Table 4: Summary of the experimental setup cases. See the text paragraph for the abbreviation definitions.

| No | Data | Type      | Proj    | Sex    | Laterality | Age       |
|----|------|-----------|---------|--------|------------|-----------|
| 1  | RDR  | Patient   | A       | Both   | All        | All       |
| 2  | ACR  | Patient   | A       | Both   | All        | All       |
| 3  | RDR  | Per image | A       | Both   | All        | All       |
| 4  | ACR  | Per image | A       | Both   | All        | All       |
| 5  | RDR  | Per image | A Vs B  | Both   | All        | All       |
| 6  | RDR  | Per image | A Vs AB | Both   | All        | All       |
| 7  | RDR  | Patient   | A       | M Vs F | All        | All       |
| 8  | RDR  | Per image | A       | Both   | L Vs R     | All       |
| 9  | RDR  | Patient   | A       | Both   | All        | $\leq 60$ |

The metrics used for the classification evaluation are based on the F1 score:(Sensitivity, Specificity, Positive Predictive Values, Negative Predicted Values, and Accuracy). In addition, the Receiver-operating characteristic curve (ROC) is employed as a visual representation of model performance across all model thresholds. Moreover, the Area Under the Curve (AUC) is measured for each ROC curve. In the following equations, we define the equation of each metric [34] :

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

$$PPV \text{ (precision)} = \frac{TP}{TP + FP}, \quad (2)$$

$$NPV = \frac{TN}{TN + FN}, \quad (3)$$

$$Sensitivity \text{ (recall)} = \frac{TP}{TP + FN}, \quad (4)$$

$$Specificity = \frac{TN}{TN + FP} \quad (5)$$

$$F1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (6)$$

$$(7)$$

where,  $TP$  = True Positive,  $TN$  = True Negative,  $FN$  = False Negative, and  $FP$  = False Positive. The following equations [34] are used for the fairness evaluation:

$$DI = \frac{P(R = 1 | A = \text{unprivileged})}{P(R = 1 | A = \text{privileged})}, \quad (8)$$

$$EOD_1 = P(R = 1 | Y = 1, A = \text{unprivileged}) - P(R = 1 | Y = 1, A = \text{privileged}), \quad (9)$$

$$EOD_0 = P(R = 0 | Y = 0, A = \text{unprivileged}) - P(R = 0 | Y = 0, A = \text{privileged}), \quad (10)$$

where unprivileged is the monitor group, privileged is the reference group and  $P(R = \text{outcome} | Y = \text{label}, A = \text{group})$  represents the probability  $P$  to get an outcome  $R$  for a label  $Y$  and a group  $A$ . In this work  $EOD_1$  when  $Y = 1$  and  $EOD_0$  when  $Y = 0$ .

To evaluate the explainability GradCam [17] were used in all the RFP (using M1-Referral model). Explainability helps clinicians, creators, patients, and regulatory bodies to understand the model errors. Note that we present only a few representative cases in this work.

## 6 Results and analysis

This section presents the results only using the local data (external clinical validation).

### 6.1 Patient referral classification

*Patient RD Reference (RDR).* As depicted in Table 4 experiment one, the images analyzed are macula-centered (projection A), both sexes, and the non-gradable images were removed. The dataset included 797 patients (743 classified as Non-Referable Diabetic Retinopathy (NRDR) and 54 as Referable Diabetic Retinopathy (RDR)). Table 5 presents a detailed classification report. The F1 score; (sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy) of the presented model here, is as follows: 98; (91, 96, 64, 99, 96). In comparison, the EyeArt system’s performance is 86; (98, 76, 23, 100, 77). These results indicate that this model significantly outperforms EyeArt, particularly in specificity, PPV, F1 score, and overall accuracy, suggesting that the model effectively identifies RDR cases while maintaining a lower rate of false positives.

*All-cause reference per patient.* This experiment is defined as number two in Table 4. The most important difference from experiment one is that we include non-gradable images (R6) in the referral class. Table 5 presents a classification report of All-Cause Reference (ACR) with 1046 patients (743 NRDR, 303 RDR). The AI systems demonstrated the following classification performance, F1;(sensitivity, specificity, PPV, NPV, accuracy): RASIS-RD (Ours), 90;(89, 86, 72, 95, 87); Eyeart 85;(93, 76, 61, 96, 81). RAIS-DR demonstrates superior performance in F1 score, specificity, PPV, and accuracy, exceeding EyeArt by 5%, 10%, 11%, and 6%, respectively. Although EyeArt has a slight edge in sensitivity, our model has higher accuracy and PPV make it more robust in clinical applications where reducing false positives and maintaining overall accuracy are key for effective patient management.

Figure 2 presents ROC curves and AUC values for our models, ACR (AUC=0.93) and DRD (AUC=0.97), using the external validation dataset. Additionally, individual points ( $\star$ ) represent the True Positive Rate (TPR) and False Positive Rate (FPR) of the EyeArt classification system. A key advantage of a locally trained model is the ability to access and adjust all relevant threshold values, providing control over the AUC. Clinical requirements exhibit significant variability across different medical applications, patient populations, and potential consequences of diagnostic errors. Therefore, maintaining precise control over AI-based systems is paramount for enhancing their efficacy, ensuring fairness, and optimizing patient outcomes.

### 6.2 Image referral classification

*RDR per image* classification metrics are defined as follows F1;(sensitivity, specificity, PPV, NPV, accuracy): RAIS-DR, 98;(89, 96, 61, 99, 96); Eyeart 85;(98, 74, 19, 100, 75). The proposed model shows higher values for F1 score, specificity, PPV, and accuracy, by 13%, 12%, 42%, and 21%, respectively. However, EyeArt has a higher sensitivity by 9% and a slightly higher NPV 1%. Table 6 lists the classification metrics per image



Table 5: Reference classification report per patient with local data, for both cases All-Cause Reference (ACR) and Reference DR RDR. The proposed system is the RASI-DR.

| <b>DRD</b>   |            |            |           |            |               |               |                |               |
|--------------|------------|------------|-----------|------------|---------------|---------------|----------------|---------------|
| <b>Model</b> | <b>TN</b>  | <b>FP</b>  | <b>FN</b> | <b>TP</b>  | <b>PPV(%)</b> | <b>Sen(%)</b> | <b>Spec(%)</b> | <b>NPV(%)</b> |
| Proposed     | <b>715</b> | <b>28</b>  | 5         | 49         | <b>64</b>     | 91            | <b>96</b>      | 99            |
| Eyearth      | 562        | 181        | <b>1</b>  | <b>53</b>  | 23            | <b>98</b>     | 76             | <b>100</b>    |
| <b>ACR</b>   |            |            |           |            |               |               |                |               |
| <b>Model</b> | <b>TN</b>  | <b>FP</b>  | <b>FN</b> | <b>TP</b>  | <b>PPV(%)</b> | <b>Sen(%)</b> | <b>Spe(%)</b>  | <b>NPV(%)</b> |
| Proposed     | <b>637</b> | <b>106</b> | 33        | 270        | <b>72</b>     | 89            | <b>86</b>      | <b>95</b>     |
| Eyearth      | 562        | 181        | <b>21</b> | <b>282</b> | 61            | <b>93</b>     | 76             | 96            |

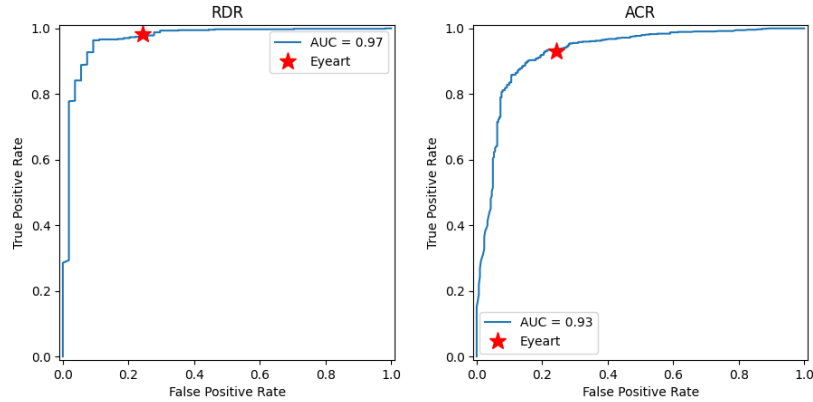


Figure 2: Left DRD ROC curve and the individual point of Eyeart system, right ACR ROC curve and the individual point of Eyeart system.

for RDR for the proposed system and the Eyeart system. Overall, this performance profile suggests that the proposed model better suited for contexts where specificity and accurate classification of true negatives are prioritized.

*ACR per image* classification following the same convention F1 score;(sensitivity, specificity, PPV, NPV, accuracy) our results are the following: Proposed, 92;(87, 89, 69, 96, 88); Eyeart 84;(93, 74, 51, 97, 78). The proposed model has higher values for F1 score, specificity, PPV, and accuracy, by 8%, 15%, 18%, and 10%, respectively, and lower values for sensitivity by 6% and NPV 1%. Table 6 presents the classification report for ACR for both the proposed model and EyeArt.

Table 6: Reference per image classification with ACR and DRD per image.

| <b>RDR</b>   |             |            |           |            |               |               |                |               |
|--------------|-------------|------------|-----------|------------|---------------|---------------|----------------|---------------|
| <b>Model</b> | <b>TN</b>   | <b>FP</b>  | <b>FN</b> | <b>TP</b>  | <b>PPV(%)</b> | <b>Sen(%)</b> | <b>Spec(%)</b> | <b>NPV(%)</b> |
| Proposed     | <b>1550</b> | <b>58</b>  | 11        | 89         | <b>61</b>     | 89            | <b>96</b>      | 99            |
| Eyearth      | 1186        | 422        | <b>2</b>  | <b>98</b>  | 19            | <b>98</b>     | 74             | <b>100</b>    |
| <b>ACR</b>   |             |            |           |            |               |               |                |               |
| <b>Model</b> | <b>TN</b>   | <b>FP</b>  | <b>FN</b> | <b>TP</b>  | <b>PPV(%)</b> | <b>Sen(%)</b> | <b>Spe(%)</b>  | <b>NPV(%)</b> |
| Proposed     | <b>1428</b> | <b>180</b> | 60        | 410        | <b>69</b>     | 87            | <b>89</b>      | 96            |
| Eyearth      | 1186        | 422        | <b>33</b> | <b>437</b> | 51            | <b>93</b>     | 74             | <b>97</b>     |

The proposed classification model demonstrates strong performance and an excellent balance across various metrics, particularly excelling in F1 score, specificity, PPV, and overall accuracy, although it shows room

for improvement in sensitivity and NPV. As mentioned, one way to improve sensitivity is to adjust the classification threshold or use data-centric techniques.

### 6.3 Fairness evaluation

In the following section, we present the results of fairness metrics. In this case, we present the results with the local data per image regarding our AI-based system. As mentioned, we compare projections A (macula-centered) and B (optic nerve-centered), sex, laterality, and age (see Table 4 experiments from five to nine). Table 7 compares the  $DI$ ,  $EOD_1$ ,  $EOD_0$ . According to the results,  $0.984 < DI < 1.031$ , suggests the absence of adverse impact, beyond the limitations of the four-fifths rule such as a small sample size, or a simplistic threshold [35]. Concerning the  $EOD_0$  and  $EOD_1$ , the results are according to the DI; the EOD values are closer to zero *i.e.* our models are fair with the different subgroups; in addition, the same subgroups have more significant values. These evaluations were important due to some convolutional models could predict the sex or age from an RFP [36, 37]. In addition, it’s important to examine other equity factors, such as long-term outcomes, or systemic barriers, that might still exist.

Table 7: Fairness metrics, Disparate Impact (DI), and Equal Opportunity Difference (EOD) when  $Y = 1$  and when  $Y = 0$ . Both metrics suggest the absence of adverse impact.

| DRD        |             |              |            |         |         |         |
|------------|-------------|--------------|------------|---------|---------|---------|
| Feature    | Type        | unprivileged | privileged | $DI$    | $EOD_0$ | $EOD_1$ |
| Projection | Per Image   | B            | A          | 0.997   | -0.005  | -0.004  |
| Projection | Per Image   | A            | AB         | 1.001   | 0.003   | 0.002   |
| Laterality | Per Image   | Left         | Right      | 1.008   | 0.003   | 0.061   |
| Sex        | Per Patient | Male         | Female     | 0.984   | -0.008  | -0.0204 |
| Age        | Per Patient | <60          | $\geq 60$  | 1.031   | -0.003  | 0.149   |
| ACR        |             |              |            |         |         |         |
| Feature    | Type        | unprivileged | privileged | $DI$    | $EOD_0$ | $EOD_1$ |
| Projection | Per Image   | B            | A          | 1.00004 | -0.012  | 0.014   |
| Projection | Per Image   | A            | AB         | 0.999   | 0.006   | -0.007  |
| Laterality | Per Image   | Right        | Left       | 0.9907  | 0.003   | -0.056  |
| Sex        | Per Patient | Male         | Female     | 0.938   | -0.026  | -0.047  |
| Age        | Per Patient | <60          | $\geq 60$  | 1.066   | 0.001   | -0.047  |

### 6.4 Explainability

One advantage of creating a model is that the explanation can be performed. Figure 3 presents a confusion matrix with image heatmaps. Two examples of True positive cases (True: RDR, Pred: RDR), show that the feature maps highlight the hemorrhages, microvascular abnormalities, and exudates that are typical characteristics of R3 (Severe non-proliferative DR). The false positive cases (True: NRDR, Pred: RDR) highlight LASER lesion areas, exudates, and anatomical deformations. In true negative cases (True:NRDR, Pred:NRDR), the heatmaps indicate that the model focuses mainly on the arcades and fovea area. False negative RFP cases (True:RDR, Pred:NRDR) occur when the model predicts R2, but the actual value is R3.

### 6.5 Deploy, monitoring, and exposure to external review

The RAIS-DR has been deployed at <https://gitlab.com/inteligencia-gubernamental-jalisco/jalisco-retinopathy> with a Responsible AI license. The system utilizes a REST-API with the AI-based system using in Docker container, which is ready to use in the cloud serverless. The system allows the tester users to upload several images and view each model’s (quality models and reference and DR-grade models) results (class and confidence interval). In addition, expert users can suggest alternative quality levels and DR grades to monitor the RAIS-DR performance.

Regarding external review, this project has been submitted at various stages to conferences [38, 21, 39], research papers [40, 18, 41], and international committees such as GPAI [42] and UNESCO’s Top-100 ( where it selected in the top-10). These practices increased the transparency of the RAIS-DR and yielded feedback.

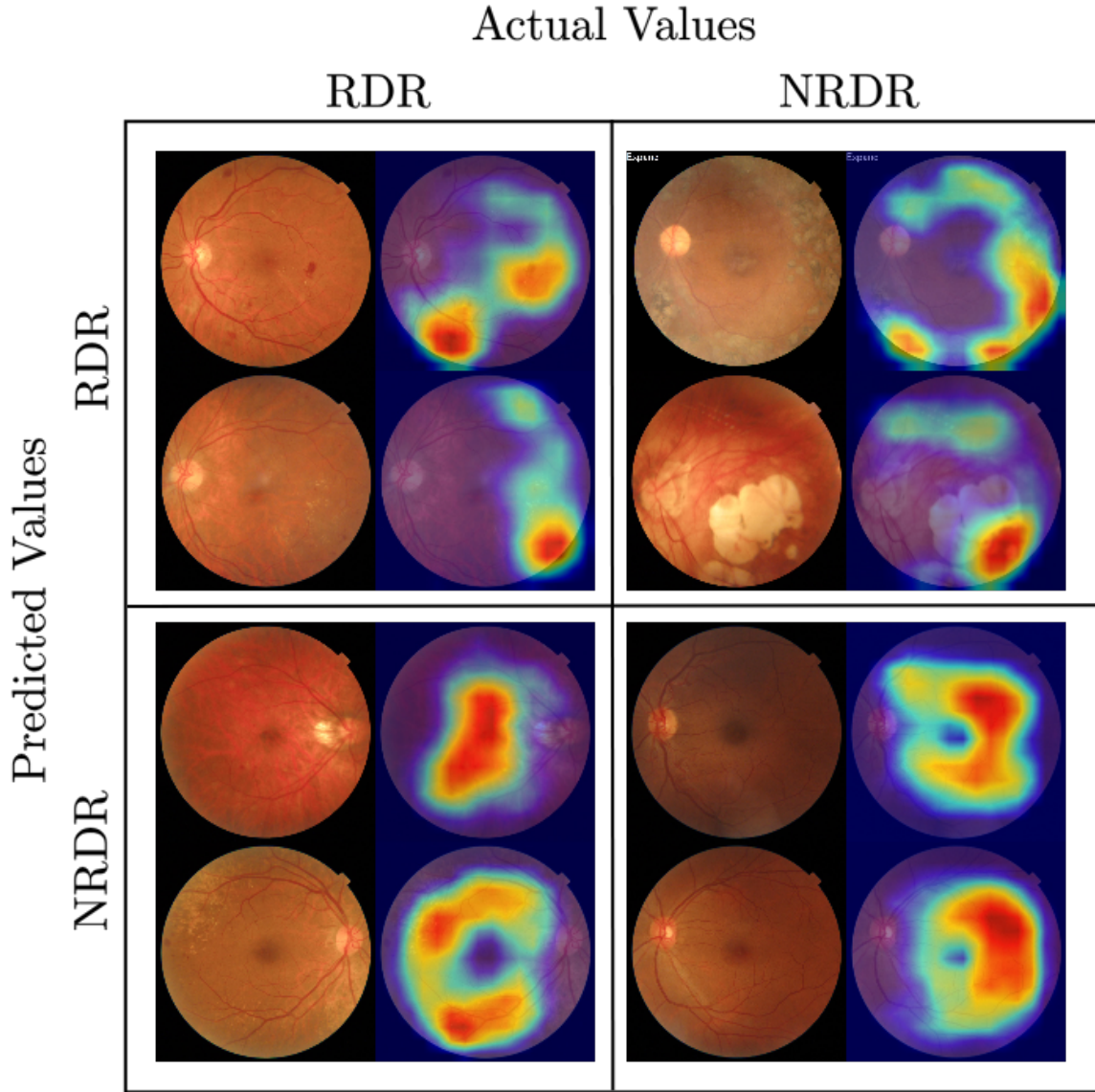


Figure 3: Confusion matrix and feature maps using GradCam [17]. It is possible to see the most important features of the model (M1-Referral) in each case.

## 6.6 Model limitations

The model cannot detect other retinal diseases, such as glaucoma or macular age degeneration. Moreover, image artifacts, low-quality images, LASER-scarring, and unusual anatomical changes in the retina will significantly reduce the confidence/performance of DR classification models.

## 7 Conclusions

This study introduced a novel Responsible AI-based System for Diabetic Retinopathy Screening (RAIS-DR) in a clinical environment, using a separate local dataset comprising approximately 1,000 patients. The here presented approach is grounded in ethical principles and specific actions in the AI lifecycle. Validation in a clinical environment underscores the practicality and real-world application of this research, enhancing

the typical AI lifecycle with responsible actions to mitigate *bias* and assess risks effectively. Key features of RAIS-DR include preprocessing techniques to prevent geometric distortions, reduce background noise, evaluate image quality during acquisition, and incorporate external evaluations throughout the AI cycle.

The RAIS-DR, was numerically compared with Eyeart, demonstrating superior specificity and effectively reducing false positives. It also exhibited a higher Positive Predictive Value (PPV) than Eyeart, indicating increased reliability when classifying instances as positive. The system outperformed Eyeart, accurately identifying both positive and negative cases with greater precision. Although the proposed model showed lower sensitivity, this presents an opportunity to improve further reduce false negatives. The model’s AUC curves indicate the potential to adjust thresholds to address the false-negative rate if necessary. Additionally, fairness metrics confirm that RAIS-DR performs equitably across different subgroups. Using home made model RAIS-DR offers additional advantages, including customization and full access to the model, thus facilitating better understanding explainability and continuous improvements.

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## References

- [1] World Health Organization
- [2] Ana Basto-Abreu, Nancy López-Olmedo, Rosalba Rojas-Martínez, Carlos A Aguilar-Salinas, Grea L Moreno-Banda, Martha Carnalla, Juan A Rivera, Martin Romero-Martinez, Simón Barquera, and Tonatiuh Barrientos-Gutiérrez. Prevalencia de prediabetes y diabetes en México: Ensanut 2022. *Salud publica de Mexico*, 65:s163–s168, 2023.
- [3] Riccardo Cheloni and Gandolum healthcare settings in Australia. Global prevalence of diabetic retinopathy: protocol for a systematic review and meta-analysis. *Scientific Reports*, 11(1):15808, 2021.
- [4] Martina Kropp, Olga Golubnitschaja, Alena Mazurakova, Lenka Koklesova, Nafiseh Sargheini, Trong-Tin Kevin Steve Vo, Eline de Clerck, Jiri Polivka Jr, Pavel Potuznik, Jiri Polivka, et al. Diabetic retinopathy as the leading cause of blindness and early predictor of cascading complications—risks and mitigation. *Epma Journal*, 14(1):21–42, 2023.
- [5] José Tomás Arenas-Cavalli, Ignacio Abarca, Maximiliano Rojas-Contreras, Fernando Bernuy, and Rodrigo Donoso. Clinical validation of an artificial intelligence-based diabetic retinopathy screening tool for a national health system. *Eye*, 36(1):78–85, 2022.
- [6] Varun Gulshan, Lily Peng, Marc Coram, Martin C Stumpe, Derek Wu, Arunachalam Narayanaswamy, Subhashini Venugopalan, Kasumi Widner, Tom Madams, Jorge Cuadros, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *Jama*, 316(22):2402–2410, 2016.
- [7] Martin J Willemink, Wojciech A Koszek, Cailin Hardell, Jie Wu, Dominik Fleischmann, Hugh Harvey, Les R Folio, Ronald M Summers, Daniel L Rubin, and Matthew P Lungren. Preparing medical imaging data for machine learning. *Radiology*, 295(1):4–15, 2020.
- [8] Eric J. Topol. High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine*, 25(1):44–56, January 2019.
- [9] Emma Beede, Elizabeth Baylor, Fred Hersch, Anna Iurchenko, Lauren Wilcox, Paisan Ruamviboonsuk, and Laura M Vardoulakis. A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy. In *Proceedings of the 2020 CHI conference on human factors in computing systems*, pages 1–12, 2020.
- [10] Michael Roberts, Derek Driggs, Matthew Thorpe, Julian Gilbey, Michael Yeung, Stephan Ursprung, Angelica I Aviles-Rivero, Christian Etmann, Cathal McCague, Lucian Beer, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for covid-19 using chest radiographs and ct scans. *Nature Machine Intelligence*, 3(3):199–217, 2021.

- [11] Alex J DeGrave, Joseph D Janizek, and Su-In Lee. Ai for radiographic covid-19 detection selects shortcuts over signal. *Nature Machine Intelligence*, 3(7):610–619, 2021.
- [12] Jessica Fjeld, Nele Achten, Hannah Hilligoss, Adam Nagy, and Madhulika Srikumar. Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for ai. *Berkman Klein Center Research Publication*, (2020-1), 2020.
- [13] EyePACS. Diabetic Retinopathy Detection competition. <https://www.kaggle.com/c/diabetic-retinopathy-detection/>, 2015. [accessed 04-September-2019].
- [14] Eye Clinic. Diabetic Retinopathy Segmentation and Grading Challenge. <https://idrid.grand-challenge.org/Data/>, 2012. [accessed 04-September-2019].
- [15] Decenci re. The Messidor database. <http://www.adcis.net/en/third-party/messidor/>, 2014.
- [16] Gam ez Karina, Mill n Yuribia, Reynoso Beatriz, Rodarte Alina, and Cort s Julieta. Gu a de pr ctica cl nica: Detecci n de retinopat a diab tica en primer nivel de atenci n. Technical report, Instituto Mexicano del Seguro Social, 2015.
- [17] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, pages 618–626, 2017.
- [18] German Pinedo-Diaz, Susana Ortega-Cisneros, Eduardo Ulises Moya-Sanchez, Jorge Rivera, Pedro Mejia-Alvarez, Francisco J Rodriguez-Navarrete, and Abraham Sanchez. Suitability classification of retinal fundus images for diabetic retinopathy using deep learning. *Electronics*, 11(16):2564, 2022.
- [19] Jiawen Lin, Lun Yu, Qian Weng, and Xianghan Zheng. Retinal image quality assessment for diabetic retinopathy screening: A survey. *Multimedia Tools and Applications*, 79:16173–16199, 2020.
- [20] Edward Korot, Zeyu Guan, Daniel Ferraz, Siegfried K Wagner, Gongyu Zhang, Xiaoxuan Liu, Livia Faes, Nikolas Pontikos, Samuel G Finlayson, Hagar Khalid, et al. Code-free deep learning for multi-modality medical image classification. *Nature Machine Intelligence*, 3(4):288–298, 2021.
- [21] Gaspar Gonz lez Briceno, Abraham S nchez, E Ulises Moya S nchez, Susana Ortega Cisneros, Germ n Pinedo, Mario S Garc a Contreras, and Beatriz Alvarado Castillo. Automatic cropping of retinal fundus photographs using convolutional neural networks. *Res. Comput. Sci*, 149:161–167, 2020.
- [22] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier, 2016.
- [23] Samuel Dodge and Lina Karam. Understanding how image quality affects deep neural networks. In *2016 eighth international conference on quality of multimedia experience (QoMEX)*, pages 1–6. IEEE, 2016.
- [24] Jane Scheetz, Dilara Koca, Myra McGuinness, Edith Holloway, Zachary Tan, Zhuoting Zhu, Rod O’day, Sukhpal Sandhu, Richard J MacIsaac, Chris Gilfillan, et al. Real-world artificial intelligence-based opportunistic screening for diabetic retinopathy in endocrinology and indigenous healthcare settings in australia. *Scientific Reports*, 11(1):15808, 2021.
- [25] John Mongan, Linda Moy, and Charles E Kahn Jr. Checklist for artificial intelligence in medical imaging (claim): a guide for authors and reviewers, 2020.
- [26] Robert F Wolff, Karel GM Moons, Richard D Riley, Penny F Whiting, Marie Westwood, Gary S Collins, Johannes B Reitsma, Jos Kleijnen, Sue Mallett, and PROBAST Group†. Probast: a tool to assess the risk of bias and applicability of prediction model studies. *Annals of internal medicine*, 170(1):51–58, 2019.
- [27] Taridzo Chomutare, Miguel Tejedor, Therese Olsen Svenning, Luis Marco-Ruiz, Maryam Tayefi, Karianne Lind, Fred Godtliebsen, Anne Moen, Leila Ismail, Alexandra Makhlysheva, et al. Artificial intelligence implementation in healthcare: a theory-based scoping review of barriers and facilitators. *International Journal of Environmental Research and Public Health*, 19(23):16359, 2022.
- [28] OECD. Advancing accountability in ai, 2023.
- [29] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
- [30] Andrew G Howard. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*, 2017.

- [31] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in neural information processing systems*, pages 91–99, 2015.
- [32] Mingxing Tan and QV Le. Efficientnetv2: Smaller models and faster training. arxiv 2021. *arXiv preprint arXiv:2104.00298*.
- [33] Luis Tejerina, Ricardo Perez-Cuevas, and Jorge García-Castro. Implementación de inteligencia artificial para mejorar la atención de la retinopatía diabética en el estado de jalisco, méxico, 2023.
- [34] Simon Caton and Christian Haas. Fairness in machine learning: A survey. *ACM Computing Surveys*, 56(7):1–38, 2024.
- [35] Manish Raghavan and Pauline T Kim. Limitations of the "four-fifths rule" and statistical parity tests for measuring fairness. *Geo. L. Tech. Rev.*, 8:93, 2024.
- [36] Edward Korot, Nikolas Pontikos, Xiaoxuan Liu, Siegfried K Wagner, Livia Faes, Josef Huemer, Konstantinos Balaskas, Alastair K Denniston, Anthony Khawaja, and Pearse A Keane. Predicting sex from retinal fundus photographs using automated deep learning. *Scientific reports*, 11(1):10286, 2021.
- [37] Nele Gerrits, Bart Elen, Toon Van Craenendonck, Danai Triantafyllidou, Ioannis N Petropoulos, Rayaz A Malik, and Patrick De Boever. Age and sex affect deep learning prediction of cardiometabolic risk factors from retinal images. *Scientific reports*, 10(1):9432, 2020.
- [38] Abraham Sanchez, Eduardo Ulises Moya, Raul Nanclares, Alexander Quevedo, Jorge Martinez, and Gaspar Gonzalez. Deep Learning Models for Diabetic Retinopathy Screening Program. In *LatinX in AI at Neural Information Processing Systems Conference 2019*, number NeurIPS, pages 3–5. Journal of LatinX in AI Research, dec 2019.
- [39] Pedro de J Bermejo-Guerrero, Abraham Sánchez, E Ulises Moya-Sánchez, and Ulises Cortés. Etiquetado, clasificación y análisis de calidad de imagen para detección de retinopatía diabética usando modelos convolucionales profundos.
- [40] Gaspar Gonzalez-Briceno, Abraham Sanchez, Susana Ortega-Cisneros, Mario Salvador Garcia Contreras, German Alonso Pinedo Diaz, and E Ulises Moya-Sanchez. Artificial intelligence-based referral system for patients with diabetic retinopathy. *Computer*, 53(10):77–87, 2020.
- [41] Ulises Cortés, Atia Cortés, Cristian Barrué, Abraham Sánchez, E Ulises Moya-Sánchez, and Dario Garcia-Gasulla. To be fair or not to be: using ai for the good of citizens. *IEEE Technology and Society Magazine*, 40(1):55–70, 2021.
- [42] GPAI and Future Society. Areas for future action in the responsible AI ecosystem . Technical report, GPAI, 12 2020.